

Smart FreshChain

Final Solution Architecture and Design

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1. Executive Summary and Final Scope

Smart FreshChain is an event-driven SAP BTP solution for refrigerated retail operations. It simulates and ingests cold-chain sensor data, persists readings and master data in SAP HANA Cloud, applies CAP-based rules and SAP AI Core ML models, and surfaces actionable alerts in a Fiori Elements application. The design is intentionally local-first so development can start immediately while the BTP subaccount and entitlements are finalised.

1.1 Final Target Capability

- Monitor temperature, humidity, CO2, O2, light and operational door/defrost/fault events per store and fridge zone.
- Persist raw readings, rolling aggregates, model predictions, alerts and staff actions in a CDS-aligned SAP HANA Cloud schema.
- Detect threshold breaches immediately using a CAP rule engine and later improve detection using SAP AI Core anomaly models.
- Predict shelf-life risk and recommended actions from environmental summaries plus product and batch metadata.
- Show store managers and staff a Fiori Elements dashboard with zone status, active alerts, risk evidence and action workflow.
- Synchronise product and batch master data from S/4HANA using SAP Integration Suite as a should-have target capability.
- Enable a later SAP Generative AI Hub copilot that answers grounded questions using live CAP data and approved tool actions.

1.2 Scope Removed or Deferred

Area	Decision	Rationale
Visual spoilage / cameras	Removed from final scope	The current project focuses on environmental and operational signals only. This avoids camera hardware, image storage, privacy and model-labeling complexity.
Physical IoT devices	Deferred	The hackathon demo runs on a simulator with realistic scenario generation. Real hardware can be introduced once ingestion contracts are stable.
POS closed loop	Out of scope for hackathon	Product backlog explicitly treats POS integration as won-not-have. Manual action logging is enough for the demo.
Gen-AI in Week 1	Deferred but retained as target design	Week 1 should not block on Gen-AI; ML and rule-based alerting remain the immediate intelligence path.
PostgreSQL	Replaced by HANA Cloud	The current design aligns CAP persistence with SAP HANA Cloud and HDI containers.

1.3 Business Outcomes

Outcome	How the Architecture Supports It
Real-time visibility	Sensor readings flow from simulator or future edge devices into Event Mesh/CAP and are visible by zone in Fiori.
Earlier intervention	Rules and ML score rolling windows and create alerts before stock loss escalates.
Operational actionability	Alerts include severity, evidence, and recommended actions such as inspect fridge, rotate stock or markdown at-risk batch.
Audit and learning loop	Staff actions, model version, prediction confidence and outcomes are persisted for traceability and future retraining.
Enterprise integration	S/4HANA product and batch master data can be introduced through Integration Suite without changing the CAP core model.

2. Target Architecture Diagrams

The following rendered diagrams show the final project architecture and complete end-to-end data path. The local development lane is included so the team can start building before the SAP BTP subaccount is fully available.

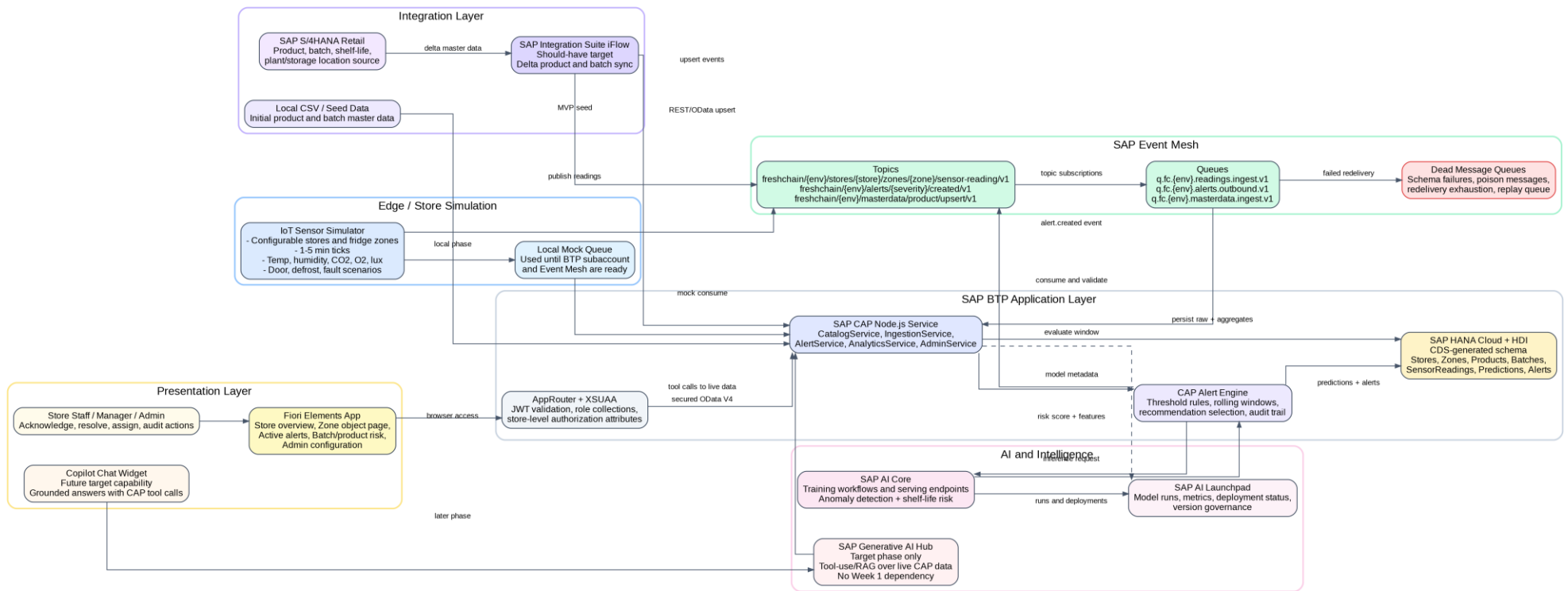


Figure 1: Final target architecture across edge simulation, SAP Event Mesh, CAP, SAP HANA Cloud, SAP AI Core, Fiori Elements and Integration Suite.

2.1 End-to-End Data Flow

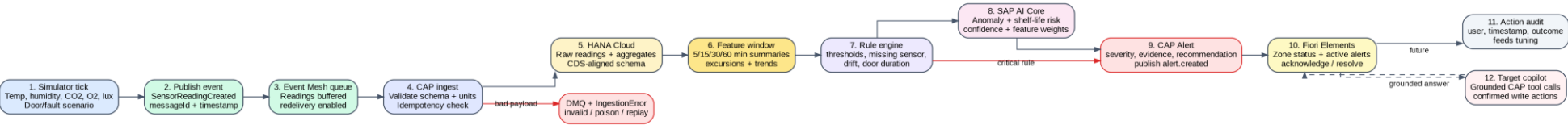


Figure 2: Complete sensor-to-alert-to-action data flow, including fallback and dead-message handling.

2.2 Deployment View

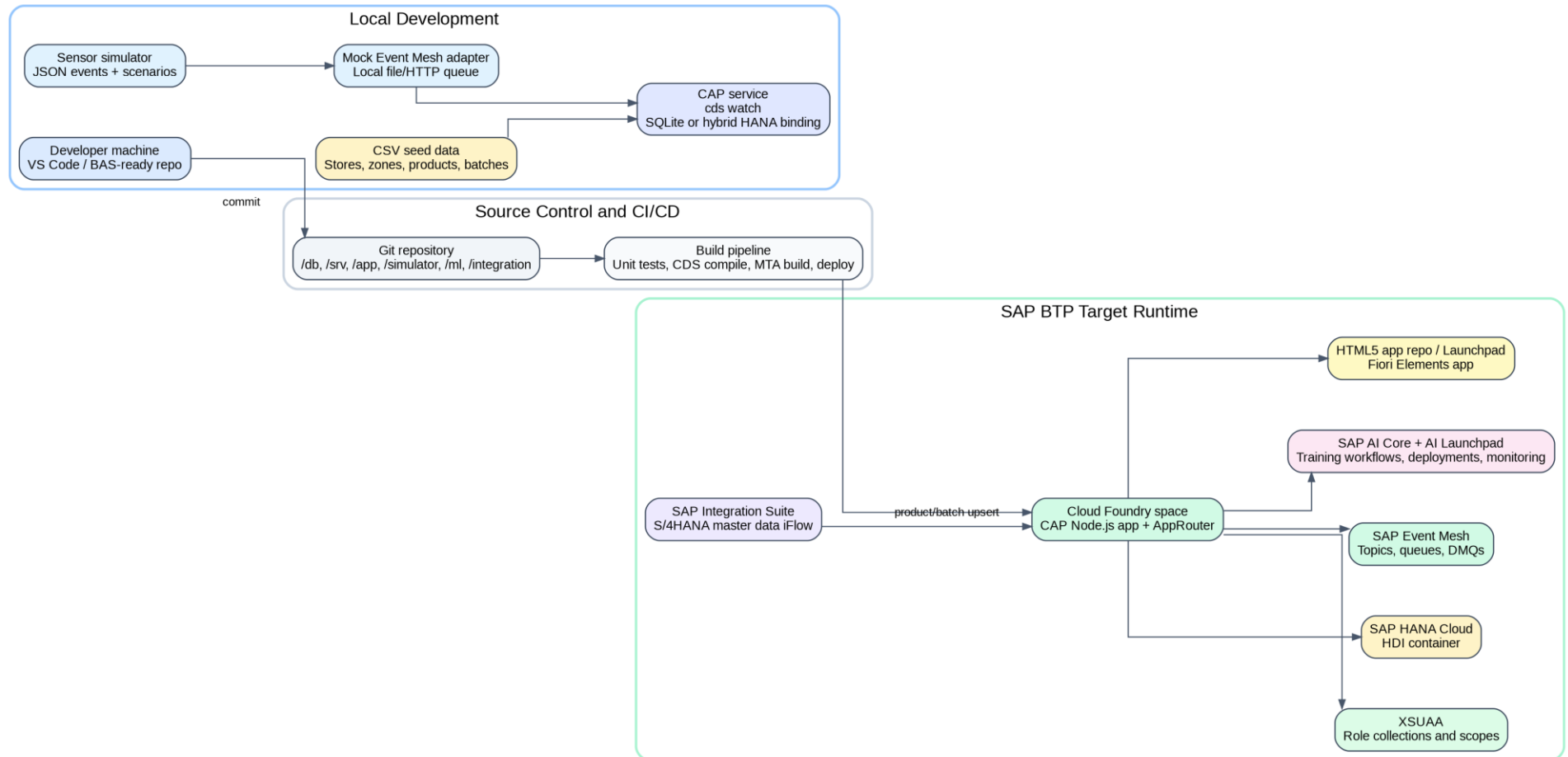


Figure 3: Local-first development path and SAP BTP target runtime view.

3. Key Architecture Decisions

ID	Decision	Implication
AD-001	Use a simulator as the ingestion source for the demo.	No hardware dependency. The message contract is the integration contract for future sensors.
AD-002	Use SAP HANA Cloud/HDI for the target database, not PostgreSQL.	CDS remains the source of truth. HANA deployment artefacts are generated and deployed through HDI.
AD-003	Use Event Mesh for BTP target ingestion and a mock queue locally.	Local development can start now; switching to Event Mesh is an adapter change, not a domain-model change.
AD-004	Use Fiori Elements over CAP OData V4 as the primary UX.	Fastest path to list reports, object pages, actions and role-based business UI.
AD-005	Keep a deterministic CAP rule engine even after AI Core is live.	Alerts still work if AI Core is unavailable or the model has low confidence.
AD-006	Use SAP AI Core for ML training and serving.	Provides a BTP-native model lifecycle path for anomaly detection and shelf-life risk.
AD-007	Treat SAP Generative AI Hub as a later target phase.	No Week 1 dependency, but final architecture has a safe tool-use/RAG design.
AD-008	Treat Integration Suite/S4 sync as should-have.	Local CSV seed data unblocks development; iFlow can replace seed data later.
AD-009	Remove image-based spoilage detection from the architecture.	Focus stays on environmental, operational and product/batch data.

Design Principle

The CAP CDS model, message schema and API contracts are the non-negotiable integration contracts. Once these are stable, simulator, Event Mesh, AI Core, UI and S/4HANA integration can evolve independently without causing rework.

4. Edge and Ingestion Layer

4.1 IoT Sensor Simulator Design

The simulator is the official data source for the hackathon build. It produces realistic environmental and operational events without physical sensor hardware. It should be implemented as a small Node.js or Python service that can run locally from configuration and publish either to a local mock queue or to SAP Event Mesh when credentials are available.

Component	Design
Topology	Store -> Zone -> Sensor. A store has multiple fridge/display zones. Each zone emits a periodic sensor-reading event and separate door/operational events.
Default frequency	One event per zone every 60 seconds for the demo; configurable to 1-5 minutes. Door events are emitted on state change and also reflected in the next reading.
Measurements	temperatureC, humidityPct, co2Ppm, oxygenPct, lightLux, doorOpen, batteryPct, signalStrength, scenarioCode. VOC can be added as optional future field.
Scenarios	Normal cooling, door left open, repeated door openings, compressor failure, defrost cycle, sensor drift, missing readings and noisy sensor.
Correlation	Each event includes messageId, correlationId, storeId, zoneId, sensorId, measuredAt, publishedAt and schemaVersion.
Local mode	Writes to local HTTP endpoint or JSON-lines queue consumed by CAP. This keeps the domain contract identical to Event Mesh mode.
BTP mode	Publishes to SAP Event Mesh topic using service key credentials and the same payload contract.

SensorReadingCreated payload contract

```
{  
  "schemaVersion": "1.0",
```

```

"messageId": "b5e39f1d-28a7-4a6f-8f28-0e0d00a1a111",
"correlationId": "ST001-ZN_DAIRY_01-2026-05-19T08:00:00Z",
"eventType": "SensorReadingCreated",
"storeId": "ST001",
"zoneId": "ZN_DAIRY_01",
"sensorId": "SIM_TEMP_HUM_GAS_001",
"measuredAt": "2026-05-19T08:00:00Z",
"publishedAt": "2026-05-19T08:00:01Z",
"readings": {
  "temperatureC": 4.2,
  "humidityPct": 63.5,
  "co2Ppm": 780,
  "oxygenPct": 20.7,
  "lightLux": 95,
  "doorOpen": false
},
"quality": {
  "batteryPct": 98,
  "signalStrength": -48,
  "sensorHealth": "OK"
},
"scenarioCode": "NORMAL"
}

```

4.2 Store and Zone Topology

Entity	Examples	Notes
Store	ST001, ST002	A retail location with timezone, region and operating hours.
Zone	ZN_DAIRY_01, ZN_MEAT_01, ZN_PRODUCE_01, ZN_FREEZER_01	A fridge, display cabinet, cold room section or shelf zone.
Sensor	SIM_TEMP_HUM_GAS_001	A simulated multi-sensor. Real sensors can later map 1:n to a zone.
Product/Batch link	Optional initially	Readings are zone-level in the MVP. If timeline allows, product/batch placement links alerts to inventory.

4.3 Event Mesh Topic and Queue Design

Purpose	Topic Pattern	Queue / Subscription
Sensor readings	freshchain/{env}/stores/{storeId}/zones/{zoneId}/sensor-reading/v1	q.fc.{env}.readings.ingest.v1 subscribes to all store/zone reading topics.
Door/operational events	freshchain/{env}/stores/{storeId}/zones/{zoneId}/operational-event/v1	q.fc.{env}.ops.ingest.v1 subscribes to door, defrost and maintenance events.
Alert created	freshchain/{env}/alerts/{severity}/created/v1	q.fc.{env}.alerts.outbound.v1 supports downstream notifications, demo logs and future integrations.
Product upsert	freshchain/{env}/masterdata/product/upsert/v1	q.fc.{env}.masterdata.ingest.v1 receives S/4HANA or CSV-seed product updates.
Batch upsert	freshchain/{env}/masterdata/batch/upsert/v1	q.fc.{env}.masterdata.ingest.v1 receives batch and best-before updates.
Dead messages	N/A	Each critical queue has a matching DMQ, e.g. q.fc.{env}.readings.ingest.dmq.v1.

4.4 Ingestion Error Handling and DLQ Strategy

- CAP validates schema version, required fields, timestamp drift, supported units, store/zone existence and duplicate messageId before persistence.

- Duplicate messages are ignored or upserted idempotently using sourceMessageId and measuredAt+zoneId uniqueness.
- Recoverable downstream failures use bounded retries with exponential backoff. The consumer must not endlessly retry poison payloads.
- Unrecoverable failures are written to IngestionError and routed to the matching dead-message queue with error class, payload hash and correlationId.
- A replay utility can move corrected DLQ payloads back to the ingest queue after schema or master-data issues are fixed.
- Alerting on ingestion health is required: DLQ size, queue depth, last-consumed timestamp, ingestion lag and error rate.

5. Backend and Data Layer

5.1 CAP Project Structure

Recommended repository layout

```
freshchain/
  app/
    freshchain-ui/          # Fiori Elements application
  db/
    schema.cds              # CDS domain model
    data/                   # CSV seed data for stores, zones, products, batches
  srv/
    catalog-service.cds     # OData V4 service for Fiori
    ingestion-service.cds   # Event/API contracts for ingestion
    alert-service.cds       # Alert actions and workflows
    admin-service.cds       # Threshold and simulator configuration
    handlers/               # JS handlers for ingestion, rules, AI calls
  simulator/
    config.yaml
    src/                    # Sensor simulation and publisher adapter
  ml/
    training/               # Feature preparation and model training jobs
    serving/                # Endpoint payload contracts and test harness
  integration/
    mapping/                # S4-to-CAP mapping specs and iFlow notes
  xs-security.json
  mta.yaml
  package.json
```

5.2 Core CDS Entities

Entity	Purpose	Key Fields / Relationships
Store	Retail location.	storeCode, name, region, timezone, active; composition to Zones.
Zone	Fridge/display/cold-room zone.	zoneCode, type, store, safeTempMinC, safeTempMaxC, safeHumidityRange, active.
Sensor	Physical or simulated sensor bound to a zone.	sensorId, zone, sensorType, firmwareVersion, lastSeenAt, healthStatus.
Product	SKU/master product used for shelf-life risk.	sku, name, category, subcategory, packagingType, recommendedTempMinC, recommendedTempMaxC, standardShelfLifeDays.
Batch	Product batch/lot with dates.	product, batchNumber, productionDate, packingDate, bestBeforeDate, receivedAt.
InventoryPlacement	Optional link between batch and zone.	batch, zone, placedAt, removedAt, quantity, unit. Enables product-specific alerts.
SensorReading	Raw time-series reading.	store, zone, sensor, measuredAt, temperatureC, humidityPct, co2Ppm, oxygenPct, lightLux, doorOpen, sourceMessageId.
ReadingAggregate	Derived rolling summaries.	zone, windowStart, windowEnd, tempAvg, tempMax,

Entity	Purpose	Key Fields / Relationships
		excursionMinutes, co2Slope, oxygenDrop, doorOpenSeconds.
Prediction	AI/rule prediction output.	modelName, modelVersion, zone, batch, predictionType, riskLevel, score, confidence, featureSnapshot.
Alert	Operational item requiring action.	store, zone, batch, severity, status, alertType, evidenceWindow, recommendation, prediction, createdAt.
AlertAction	Staff/manager action history.	alert, actionType, assignedTo, performedBy, comment, outcome, completedAt.
IngestionError	Failed message tracking.	sourceQueue, messageId, correlationId, errorClass, errorMessage, payloadHash, retryCount, status.
ThresholdConfig	Per-zone threshold configuration.	zoneType/productCategory thresholds, duration rule, severity mapping.

5.3 Service Surface

CAP Service	Consumers	Responsibilities
CatalogService	Fiori Elements	Expose Stores, Zones, Products, Batches, SensorReadings projections, ReadingAggregates, Alerts and AlertActions as OData V4.
IngestionService	Event Mesh consumer, local simulator	Validate and persist sensor, operational and master-data events. Internal service; not exposed to end users.
AlertService	Fiori, CAP handlers	Create alerts, acknowledge, assign, resolve, reopen, add action notes, publish alert.created events.
AnalyticsService	Fiori charts, copilot tools	Expose current zone status, risk summaries, trend windows and action KPIs.
AdminService	Admin role only	Manage thresholds, zone configuration, simulator store/zone setup and model deployment switches.
AIGatewayService	CAP Alert Engine	Wrap SAP AI Core endpoint calls, model versioning, retries, circuit-breaker and fallback metadata.

5.4 OData and REST/API Contracts

Surface	Method / Entity / Action	Notes
OData V4	Stores, Zones, ZoneStatus, Products, Batches, Alerts, AlertActions	Primary Fiori consumption. Use projections and annotations rather than exposing raw tables directly.
OData action	acknowledgeAlert(alertId), resolveAlert(alertId, outcome), assignAlert(alertId, userId)	State-changing actions require StoreStaff or StoreManager role.
OData action	triggerManualRiskEvaluation(zoneId)	Admin/StoreManager action to force model/rule evaluation for demo and testing.
Internal REST	POST /ingest/sensor-readings	Local simulator and test harness path before Event Mesh is available. Same payload as event body.
Internal REST	POST /ai/predict/anomaly, POST /ai/predict/shelf-life	Optional wrapper for AI Gateway handlers; direct user access disabled.

5.5 SAP HANA Cloud Schema, Retention and Indexing

The CDS model is the source of truth. HANA tables are generated and deployed through HDI, and CAP handlers should avoid hand-written schema drift. Time-series volume is manageable in the demo but the design anticipates production growth through partitioning, aggregation and targeted indexes.

Data Set	Retention Policy	Index / Partition Strategy
SensorReading raw	90 days for hackathon/demo; proposed production starting point 180 days, then purge or archive if a BTP archive service is approved.	Range partition by measuredAt when volume grows. Index zone_ID+measuredAt, store_ID+measuredAt, unique sourceMessageId.
ReadingAggregate	13 months.	Index zone_ID+windowStart+windowSize and store_ID+windowStart for dashboard and model training windows.

Data Set	Retention Policy	Index / Partition Strategy
Prediction	12 months or as required by model governance.	Index modelName+modelVersion+createdAt and zone_ID+createdAt.
Alert and AlertAction	24 months minimum for audit and demo evidence.	Index status+severity+createdAt and store_ID+zone_ID+status.
Product, Batch, Placement	Current plus historical batch records until business retention is agreed.	Index sku, batchNumber, bestBeforeDate, zone_ID+active placement.
IngestionError	30 days after resolved; unresolved retained until manually closed.	Index status+createdAt and sourceQueue+messageld.

5.6 Data Quality Guardrails

- Reject or quarantine values outside physical plausibility ranges, for example oxygenPct greater than 25 or negative humidity.
- Flag stale sensors when a zone has no reading for more than two expected intervals.
- Mark readings with quality flags rather than deleting them if they are useful for model robustness analysis.
- Normalize units at ingestion so all downstream logic uses Celsius, percentage humidity, ppm CO2, percentage oxygen and lux.
- Store model feature snapshots to explain why an alert was raised and to enable reproducible model evaluations.

6. AI and Intelligence Layer

6.1 Intelligence Strategy

The intelligence layer is deliberately staged. The system starts with rules because they are explainable, testable and do not require labelled anomaly data. SAP AI Core then adds anomaly detection and shelf-life risk prediction. The copilot is a later target phase and must be grounded in live CAP data rather than free-form generation.

Capability	Week 1 / Foundation	Final Target
Threshold alerting	Implemented in CAP with zone thresholds and duration windows.	Still active as deterministic fallback and safety layer.
Anomaly detection	Model contract and feature windows defined; simple model can be trained if free-tier capacity allows.	AI Core serving endpoint scores multivariate windows and detects equipment/mishandling patterns.
Shelf-life risk	Synthetic dataset and endpoint contract prepared.	Prediction uses product/batch metadata plus environmental summaries to estimate risk and days remaining.
Gen-AI copilot	Not a Week 1 dependency.	SAP Generative AI Hub tool-use/RAG over CAP APIs, with read-only responses by default and confirmation for write actions.

6.2 Dataset Preparation

Input Source	Features / Labels Produced
SensorReading	Temperature/humidity/gas/light raw values; rolling averages, maxima, minima, standard deviation, slopes and excursion minutes.
Operational events	Door-open count, total door-open seconds, defrost cycle flags, maintenance/fault flags.
Product/batch metadata	Category, packaging type, recommended temperature range, standard shelf-life days, best-before date and placement duration.
Synthetic scenarios	Ground-truth labels for normal, door-left-open, compressor fault, sensor drift, defrost and missing-data conditions.
Staff actions/outcomes	Resolved alert outcomes and comments for future retraining and threshold tuning.

6.3 Model Selection

Model	Purpose	Why This Choice
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Model	Purpose	Why This Choice
Rules	Immediate anomaly/threshold detection.	Transparent, deployable without labels, easy to demo and safe fallback.
Isolation Forest or One-Class SVM	Unsupervised anomaly detection on multivariate time-series windows.	Good baseline when normal operation data is easier to simulate than labelled failures.
Autoencoder	Later anomaly model for richer multivariate patterns.	Useful after more normal data is collected. More complex than needed for initial demo.
Gradient Boosting	Shelf-life risk and remaining-days estimate from tabular features.	Strong tabular baseline with feature importance. No image inputs required.
Survival model	Future shelf-life uncertainty and probability curves.	Useful if enough product/batch outcome labels are collected.

6.4 Evaluation Criteria

Area	Metrics
Anomaly detection	Precision, recall where labels exist, false alert rate per fridge per week, early detection rate, alert latency, manual override rate.
Shelf-life risk	MAE/RMSE for remaining shelf-life days, calibration of low/medium/high risk, high-risk hit rate, business days preserved after intervention.
System performance	Ingestion lag, AI endpoint p95 latency, queue depth, dashboard refresh time, CAP response time.
Business outcomes	Waste/shrinkage reduction, quality complaint reduction, markdown timing improvement, completed action rate.

6.5 AI Core Serving Endpoint Contract

AI Core inference request

```
POST /v1/freshchain/predict
{
  "requestId": "REQ-20260519-0001",
  "predictionType": "ANOMALY_AND_SHELF_LIFE",
  "modelVersionPolicy": "PINNED",
  "storeId": "ST001",
  "zoneId": "ZN_DAIRY_01",
  "windowStart": "2026-05-19T07:30:00Z",
  "windowEnd": "2026-05-19T08:00:00Z",
  "features": {
    "temperatureAvgC": 5.7,
    "temperatureMaxC": 9.1,
    "tempExcursionMinutesOver8C": 18,
    "humidityAvgPct": 66.1,
    "co2SlopePpmPerMin": 2.5,
    "oxygenDropPct": 0.2,
    "doorOpenSeconds": 220
  },
  "context": {
    "zoneType": "DAIRY_FRIDGE",
    "productCategory": "DAIRY",
    "batchId": "BATCH-OPTIONAL"
  }
}
```

AI Core inference response

```
{
  "predictionId": "PRED-20260519-0001",
  "modelName": "freshchain-risk",
  "modelVersion": "1.0.0",
```

```

"riskLevel": "HIGH",
"anomalyScore": 0.87,
"anomalyType": "TEMPERATURE_EXCURSION",
"remainingShelfLifeDays": 1.8,
"confidence": 0.74,
"featureContributions": [
  {"feature": "tempExcursionMinutesOver8C", "weight": 0.46},
  {"feature": "doorOpenSeconds", "weight": 0.31}
],
"recommendedActions": [
  "Inspect fridge door seal and airflow",
  "Prioritise affected dairy batches for review or markdown"
],
"fallbackUsed": false
}

```

6.6 Fallback Behaviour

- AI Core failure must not block alerting. CAP uses deterministic thresholds when the AI endpoint is unavailable or returns low confidence.
- CAP records fallbackUsed=true and modelUnavailableReason for traceability.
- The AI Gateway circuit-breaker opens after repeated endpoint failures and re-checks health periodically.
- Model versions are pinned for demo stability. Model promotion requires smoke tests against known simulator scenarios.
- Severe safety threshold breaches create alerts immediately; AI enrichment can arrive later asynchronously.

6.7 Target Gen-AI Copilot Pattern

The copilot is retained in the final solution design but explicitly excluded from the Week 1 gate. It should be implemented only after core CAP data, alerts and dashboard are stable.

Design Element	Target Behaviour
Grounding	All answers are grounded through CAP tools such as getActiveAlerts, getZoneStatus, getBatchRisk and getRecommendedActions.
RAG/tool-use	Use tool calls for live operational data. Optional RAG can index architecture/runbook documents, not raw sensor time series.
Safety	Responses must cite alert IDs, zone IDs and timestamps. The assistant must not invent products, batches or recommendations.
Actions	Read-only by default. Mutating actions such as acknowledgeAlert or createAction require explicit user confirmation.
Week 1 handling	Do not build the widget in Week 1; keep only the API/tool-contract design.

7. Presentation Layer

7.1 Fiori Elements Page Structure

Page	Users	Content / Actions
Launch / Overview	Store Staff, Store Manager, Admin	Store health cards, red/amber/green zone summary, active alerts count, last ingestion timestamp.
Store List Report	Manager, Admin	Stores with aggregated risk, open alerts, ingestion health and filter by region.
Store Object Page	Manager, Staff	Zones, recent alerts, current environment summary and links to zone details.
Zone Object Page	Manager, Staff	Current readings, trend chart, threshold status, latest AI prediction, related alerts and placement data.
Active Alerts List	Staff, Manager	Open alerts ranked by severity; actions: acknowledge, assign, resolve, add comment.
Alert Object Page	Staff, Manager	Evidence window, breach values, model version, recommendation, action history and audit trail.
Product / Batch List	Manager, Admin	SKU, category, standard shelf-life, batch dates, placement status and risk summary if

Page	Users	Content / Actions
		linked.
Admin Configuration	Admin	Thresholds, zone types, simulator configuration, model version toggle and retention controls.
Model Monitoring	Admin, Data Scientist	Model version, endpoint health, fallback rate, false alert feedback, latest training run metadata.
Copilot Widget	Future Staff/Manager	Ask grounded operational questions and request summaries after live CAP data and alerts are stable.

7.2 API Consumption Pattern

- Fiori Elements consumes CAP OData V4 projections and annotations for list reports, object pages, filters, value helps and actions.
- Avoid custom REST calls from UI unless there is a clearly non-OData interaction such as a copilot streaming endpoint.
- UI refresh can be polling-based for the demo, for example every 30 seconds. Event-driven UI push can be deferred.
- Use CAP actions for alert state changes so authorization and audit logging stay centralised.
- Expose chart-friendly projections such as ZoneStatus and ReadingAggregate instead of forcing the UI to process raw time-series data.

8. Integration Layer

8.1 S/4HANA Master Data Sync Target

For MVP and early development, product and batch data are seeded from CSV. The final should-have integration uses SAP Integration Suite to synchronise product and batch master data from S/4HANA into CAP. This keeps the core solution useful even before S/4HANA access is ready.

iFlow Step	Description
Trigger	Scheduled delta every 15 minutes for demo target; manual trigger for testing. Event-based trigger can be added if S/4HANA emits relevant business events.
Fetch delta	Call approved S/4HANA product and batch APIs or views. Exact API names must be confirmed with the S/4HANA system owner.
Transform	Map material/product, category, shelf-life, temperature recommendations, batch number and best-before date to CAP Product and Batch entities.
Validate	Check mandatory fields, date formats, duplicate keys and store/plant mapping. Invalid records go to an exception path.
Publish/upsert	Send product/batch upsert messages to Event Mesh or call CAP IngestionService REST/OData endpoint.
Error handling	Use exception subprocess, retry for transient connectivity errors, dead-message route for data-quality errors and alert Admin on repeated failures.

8.2 S/4HANA to CAP Mapping

S/4HANA Source Concept	CAP Target	Notes
Material / Product ID	Product.sku	Stable external key; store in externalProductId if format differs.
Material Description	Product.name	Language handling can be deferred for demo.
Material Group / Product Hierarchy	Product.category, Product.subcategory	Used by shelf-life and threshold logic.
Base Unit of Measure	Product.uom	Used for batch quantity display.
Total shelf life / expiration settings	Product.standardShelfLifeDays	Confirm exact S/4 source field with process owner.
Recommended storage condition	Product.recommendedTempMinC / MaxC	May come from custom attributes if not standard.
Batch / Lot number	Batch.batchNumber	External key together with product.

S/4HANA Source Concept	CAP Target	Notes
Production, packing, best-before dates	Batch.productionDate, packingDate, bestBeforeDate	Used for risk calculations and dashboard.
Plant / Storage location	Store / Zone mapping	Requires mapping table because a storage location may not equal fridge zone.

9. Security, Authorization and Observability

9.1 XSUAA Authorization Model

Role	Scope	Allowed Capabilities
StoreStaff	Operational user in assigned store(s).	View assigned store/zone status, view alerts, acknowledge alerts, add comments, mark own actions complete.
StoreManager	Manager for assigned store(s).	All StoreStaff capabilities plus assign alerts, resolve alerts, view batch risk, trigger manual evaluation and view store-level KPIs.
Admin	Technical/business administrator.	Manage stores, zones, thresholds, simulator config, model version flags, replay ingestion errors and view all stores.
DataScientist	AI operator.	View model runs, feature datasets, predictions, model metrics and training configuration; no operational state changes unless separately granted.
IntegrationUser	Technical user for Event Mesh/iFlow/CAP system calls.	Ingest readings, product/batch upserts and publish internal events only. No Fiori access.

Use CAP @requires and @restrict annotations for entity/action protection. For store-level isolation, model a user attribute such as storeIds and apply CAP handlers or CDS restrictions so staff and managers only see their assigned stores. Technical ingestion clients use separate OAuth credentials and must not receive end-user role collections.

9.2 Observability and Operations

Area	Signals to Monitor
Ingestion	Queue depth, DLQ depth, ingestion lag, events/minute, schema failure rate, duplicate message count.
CAP service	Request latency, error rate, memory/CPU, handler failures, database connection health.
HANA Cloud	Table growth, partition health, query latency on SensorReading and Alert projections, backup status.
AI Core	Endpoint health, p95 inference latency, fallback rate, model version, training run status, prediction confidence distribution.
Fiori	Load time, OData errors, failed actions, user role mapping issues.
Integration Suite	iFlow success/failure rate, last delta sync, failed records, replay count.

9.3 Audit Requirements

- Every alert must preserve createdBy/source, severity, rule/model evidence, model version and evidence time window.
- Every staff action must preserve user, timestamp, action type, comment, previous status and new status.
- Every master-data sync batch must preserve source system, delta timestamp, processed count, failure count and correlationId.
- Every AI prediction used in an alert must preserve modelName, modelVersion, confidence, feature snapshot and fallback status.

10. Delivery Plan and Backlog Alignment

The uploaded backlog is structured around a four-week BTP demo loop: simulate sensor data, predict risk, recommend actions and present a dashboard. The final solution design keeps that intent but updates the database target to SAP

HANA Cloud and removes image analysis from scope. The backlog also shows S/4HANA integration as a should-have and POS integration as out of scope.

Phase	Backlog Items	Final Design Deliverables
Phase 1: Foundation	US-001 to US-005	Simulator, local mock queue/Event Mesh adapter, CAP scaffold, CDS model, HANA Cloud/HDI direction, raw reading persistence and queryable OData service.
Phase 2: Data and ML pipeline	US-006 to US-009	Secured OData API, alert threshold engine, synthetic dataset preparation, AI Core shelf-life model training and serving endpoint.
Phase 3: Intelligence and dashboard	US-010, US-011, US-015	AI Core anomaly detection, CAP integration of predictions, active risk status, Fiori zone overview and dashboard.
Phase 4: Action workflow and integration	US-016, US-018	Active alerts UI with acknowledge/resolve, action audit, and S/4HANA product/batch sync as should-have.
Target extension	US-012, US-013, US-014, US-017	Gen-AI copilot, action recommendations, daily summary and embedded chat widget after core data and alert flows are stable.
Removed	US-019	Image-based visual spoilage detection removed from architecture.
Not included	US-020	POS closed-loop markdown integration excluded from hackathon/final base design.

10.1 Build Order for the Team

1. Freeze the SensorReadingCreated payload and CAP CDS model before writing UI or AI code.
2. Build local simulator -> local CAP ingestion -> HANA/SQLite development persistence first.
3. Introduce Event Mesh as an adapter once service credentials exist; do not change the payload contract.
4. Implement threshold alerts before AI. This gives a reliable demo path and training data labels.
5. Add AI Core serving through AiGatewayService and keep fallback active.
6. Build Fiori pages from OData projections and CAP actions; avoid UI-side business logic.
7. Introduce S/4HANA product/batch sync only after Product and Batch entities are stable.
8. Introduce Gen-AI copilot only after live CAP APIs and alert actions are stable enough for grounding.

10.2 Definition of Done for Final Demo

Area	Done When
Simulator	A scenario can drive at least one zone from normal to high risk and back to normal using configured ticks.
Ingestion	Sensor events are persisted exactly once or idempotently ignored, and invalid messages are traceable through IngestionError/DMQ.
CAP/HANA	Stores, zones, readings, aggregates, predictions, alerts and actions are queryable through secured OData.
AI	At least one model endpoint returns risk/anomaly output and the CAP fallback works when the endpoint is unavailable.
UI	A store user can see active alerts, inspect evidence, acknowledge and resolve an alert.
Integration	CSV seed is available; S/4HANA iFlow is ready or documented as stretch with field mapping and error handling.
Security	StoreStaff, StoreManager and Admin access paths are demonstrably different.

11. Entitlement Gaps and Blockers

Service / Access	Required For	Impact If Missing	Fallback / Mitigation
SAP HANA Cloud + HDI container	Target persistence	Cannot validate final database deployment.	Use SQLite locally or hybrid binding until HANA is available; keep CDS model HANA-ready.
SAP Event Mesh	BTP event ingestion	Cannot prove target queue/topic flow.	Use local mock queue/HTTP endpoint with identical payload; switch adapter later.
XSUAA	Role-based access and AppRouter security	Fiori and CAP cannot prove target auth model.	Use mocked users locally; generate xs-security.json from CDS roles once ready.

Service / Access	Required For	Impact If Missing	Fallback / Mitigation
Cloud Foundry runtime	Deploy CAP, AppRouter and HTML5 app	Only local demo possible.	Continue local cds watch; prepare mta.yaml.
SAP AI Core + AI Launchpad	ML training, deployment and monitoring	AI scoring cannot run on target service.	Use local sklearn baseline and keep AI Gateway contract fixed.
SAP Generative AI Hub	Target copilot	Copilot cannot be demonstrated.	Defer; not needed for Week 1 or core ML path.
SAP Integration Suite	S/4HANA master data sync	No target iFlow deployment.	CSV seed data and CAP upsert API remain sufficient for MVP.
S/4HANA API access	Real product/batch master data	Only synthetic/demo products available.	Use curated CSV seed data with stable external IDs.
Destination/Connectivity	Secure S/4HANA and external service calls	iFlow/CAP cannot reach private APIs.	Mock or local CSV until network path is approved.
Application Logging / Audit Log	Operational support and audit	Reduced production observability.	Add structured logs locally and persist audit-critical events in CAP.

11.1 Biggest Blockers to Resolve First

- BTP subaccount access and entitlement confirmation, especially HANA Cloud, Event Mesh, XSUAA and AI Core.
- Agreement on the SensorReadingCreated schema and zone topology, because all other components depend on it.
- HANA Cloud vs local development profile, including whether local SQLite is acceptable until HDI is bound.
- S/4HANA product and batch access, or agreement that CSV seed data is the accepted fallback.
- AI Core free-tier capacity and service-key setup for training and serving.

12. Acceptance Test and Success Criteria

12.1 Golden Demo Scenario

- Start simulator for ST001 with zones ZN_DAIRY_01 and ZN_MEAT_01 in normal state.
- Trigger door-left-open or compressor-fault scenario on ZN_DAIRY_01.
- Sensor readings publish to Event Mesh or local mock queue and are persisted in SAP HANA Cloud through CAP.
- CAP computes rolling features and detects breach through threshold rules; AI Core enriches with anomaly score if available.
- CAP creates HIGH severity alert with evidence window and recommended action.
- Fiori Elements dashboard shows the affected zone in red and lists the active alert.
- StoreStaff acknowledges the alert; StoreManager resolves it after the simulated condition recovers.
- AlertAction audit history shows who acted, when, and what outcome was recorded.
- If AI Core is disabled, the same flow still works using rule fallback and marks fallbackUsed=true.

12.2 Success Criteria

Criterion	Target
End-to-end flow	A simulated breach creates a persisted alert visible in Fiori within 30 seconds in the demo environment.
Ingestion throughput	At least 100 events/minute in the demo with no data loss under normal conditions.
Data integrity	Duplicate message replay does not create duplicate SensorReading records.
Security	StoreStaff cannot access Admin configuration; Admin can manage thresholds; technical users cannot open Fiori.
AI fallback	Alerting continues when AI endpoint is unavailable.
Observability	Queue depth, DLQ count, ingestion errors, fallback rate and last reading timestamp are visible to Admin.
Demo narrative	Judges can understand the loop: simulate -> ingest -> predict -> recommend -> act.

13. References and Source Basis

Source artefacts reviewed for this solution design:

- FreshChain_Hackathon_Backlog.xlsx: project overview, product backlog and four-week sprint plan.
- BTP Hackathon Data Science Recommendations.pdf: environmental sensor data, operational event data, shelf-life risk and anomaly-detection recommendations.

Official SAP documentation referenced for target platform design:

Reference	URL
SAP CAP - Using SAP HANA Cloud for Production	https://cap.cloud.sap/docs/guides/databases/hana
SAP HANA Cloud - HDI Containers	https://help.sap.com/docs/hana-cloud-database/sap-hana-cloud-sap-hana-database-developer-guide-for-cloud-foundry-multitarget-applications-sap-web-ide-full-stack/sap-hdi-containers
SAP Event Mesh - Default Plan Concepts	https://help.sap.com/docs/event-mesh/event-mesh/event-mesh-default-plan-concepts
SAP Event Mesh - Manage Queue Subscriptions	https://help.sap.com/docs/event-mesh/event-mesh/manage-queue-subscriptions
SAP PubSub/Event Mesh - Configuring Dead Message Queues	https://help.pubsub.em.services.cloud.sap/Cloud/Broker-Manager/configuring-dmqs-broker-manager.htm
SAP AI Core - What Is SAP AI Core?	https://help.sap.com/docs/sap-ai-core/sap-ai-core-service-guide/what-is-sap-ai-core
SAP AI Core - Models and Scenarios	https://help.sap.com/docs/sap-ai-core/sap-ai-core-service-guide/models-and-scenarios
SAP Generative AI Hub	https://help.sap.com/docs/sap-ai-core/generative-ai/generative-ai-hub
SAP CAP - Serving Fiori UIs	https://cap.cloud.sap/docs/guides/uis/fiori
SAP CAP - Authorization	https://cap.cloud.sap/docs/guides/security/authorization
SAP CAP - Authentication and XSUAA	https://cap.cloud.sap/docs/guides/security/authentication
SAP Cloud Integration - Handle Errors Gracefully	https://help.sap.com/docs/cloud-integration/sap-cloud-integration/handle-errors-gracefully